Class I and Class II Cavities

A Full Guide



Introduction



Class I and II cavities are the most common type of cavities encountered in everyday dentistry.
Understanding their principles is essential before practical application.

<u>Your goals while drilling your cavity should be:</u>

- Every cut you make into the tooth should have a purpose.
- Establish correct outline form.
- Maintain the proper depth for each cavity type.
- Preserve sound tooth structure.
- Create walls and margins that support the future restoration.
- Preserve pulp vitality wherever possible.

What you Need for the Perfect Cavity Prep



Micromotor for low speed



High/low speed contra



Mirror + Explorer



Probe



330 carbide bur

OR



Round carbide bur



245 carbide bur for class II



Inverted diamond bur (optional)

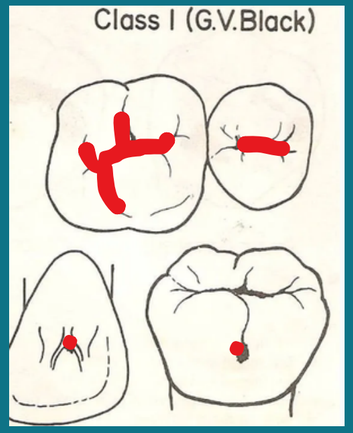


Tapered with round end yellow

Class I Cavity

To be very specific, the type of cavities being discussed in this presentation are cavities made for **COMPOSITE** restorations.

Cavities made for amalgam restorations will be explained in a separate presentation.





So, what is a Class I cavity?

According to the GV Black classification,

Class I: Cavities in pits and fissures on the occlusal surfaces of molars and premolars, buccal or lingual pits of molars, and lingual pits of maxillary incisors.

In a clinical setting, Class I cavity outline is usually determined by the **extent** and **pattern** of caries.

However, in preclinical setting, our acrylic tooth is completely sound with no caries pattern to follow.

That is why we follow specific criteria and measurements to ensure the correct outline form.





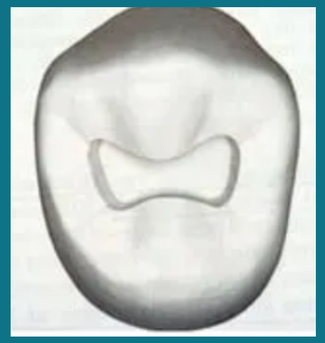
Class I Cavity Criteria

An ideal Class I cavity should:

- Be 1.5 mm deep (equal to the height of 330 bur)
- Be 1.5 mm wide
- Follow the natural grooves of the occlusal surface.
- Cut only ⅓ of the groove length
- Preserve cusps as much as possible
- Have smooth, even floor.
- Have smooth , rounded, parallel walls.
- Preserve the marginal ridge (meaning the cavity stops up to 2mm before reching the marginal ridge).







Class I Cavity Prep

Let’s prep a class I cavity in lower 6 making sure to follow all the criteria:




Class I Cavity Prep

Class II Cavity

What is a Class II Cavity?

According to GV Black classification, Class II cavities are defined as decay or restorations located on the proximal surfaces of posterior teeth - premolars and molars.
They are usually caused by food remnants and plaque build up interproximally between teeth.
This type of cavity is by far the most common type of cavity seen in everyday dentistry.

Class II cavities rarely exist in the oral cavity as JUST class II cavities, rather as class I + class II.

That is why it is more useful to always practice a combo of class I and II on your acrylic tooth to be able to mimic real life clinical situations. Even if a cavity is just a proximal box, drilling in the occlusal surface (making a class 1 cavity) is still needed with some restorations like glass ionomer to provide a mechanical lock that keeps the restoration in place.









Class II Cavity Criteria



In addition to the criteria mentioned previously for the class I part of the cavity, here are the specific measurements for the proximal box itself :

- The depth of the proximal box should be 3mm maximum (same length of 245 bur)
- The width of the proximal box should be between 1.5-2mm
- The length of the proximal box should not be more than ⅔ of the total **buccolingual** dimension of the tooth.
- Conatct with the adjacent tooth should be properly opened.
- The floor should be even and smooth
- The lingual wall of the box should be **Parallel** or **Slightly Divergent.**
- The buccal wall of the box should be **Parallel** or **Slightly Convergent.**
- The isthmus area is the joint area between the occlusal part of the cavity and the proximal box.
- The isthmus area should have a clear border with **parallel wall NOT** a **C-shape wall.**
- **Buccal cusps** should have a **Reverse S Curve.**
- **Lingual Cusps** should have **Universal Curves.**



Class II Cavity Criteria

Let’s look at some diagrams to clear up any confusions:

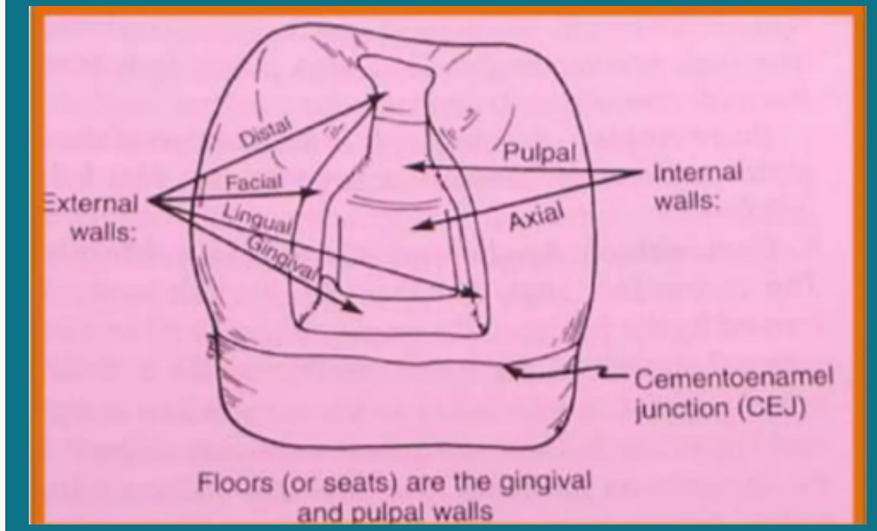


Diagram showing the correct naming of each wall in class II cavity

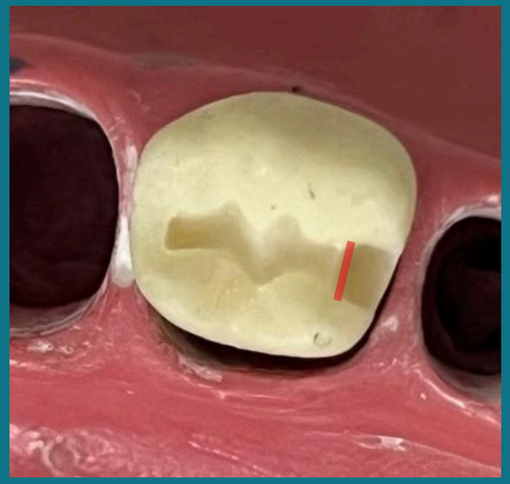


Photo showing isthmus area having a clear border and straight wall

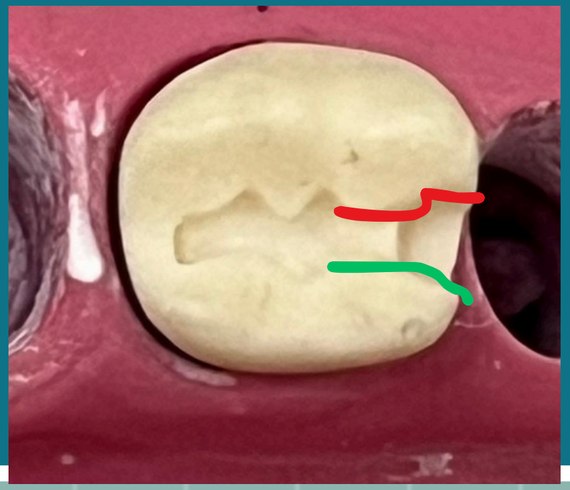


Photo showing a Mesio-occlusal cavity prep that followed frep criteria properly.
Also shown Reverse S curve in red and Universal curve in green

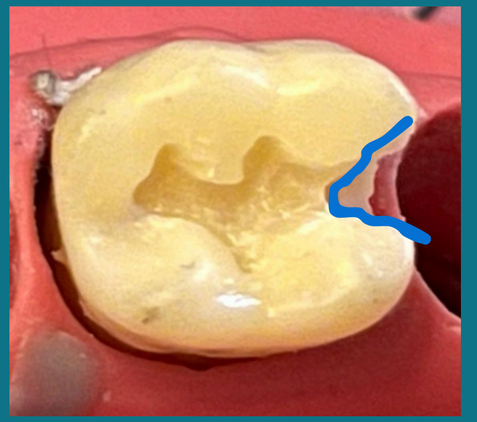


Photo shows a Mesio-occlusal cavity prep that did not follow prep criteria properly. Notice the C-Shape of the isthmus border.
What other mistakes can you spot?

Class II Cavity Prep

Lets prep a Mesio-occlusal cavity on a lower 6 following all the criteria:



Class II Cavity Prep




checking the contact

Finishing your Prep

We promise you after this step, your’e really done!

Let’s talk about finishing your prep.
This step is often skipped by students thinking that it does not matter. Let us tell you why it **does** matter:

- Removing any sharp /undermined enamel.
- Removing irregularities provides better retention of the restoration.
- Reduce stress concentration.
- Conclusion: Do NOT skip finishing if you want that A+ Prep!





How to check your work?

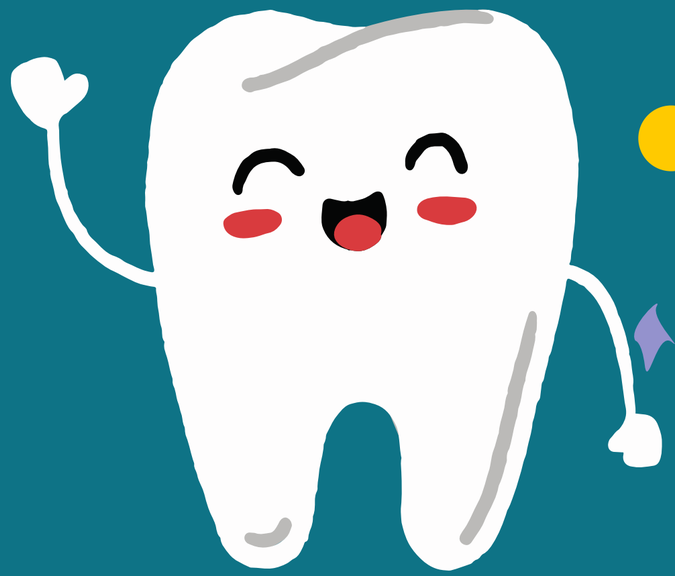


Check the MEASUREMENTS done using probe.	Check that there is no undermined enamel at any margin.	check that there is no roughness or any sharp angles
Check the measurements at different points to make sure the prep is even in all areas.	Check that floor and gingival seat are smooth and even	
check that there is no injury to adjacent tooth	Check that the grooves done are following the anatomy	
check that the contact is open properly	Check that the outline is continous with sweeping curves.	



How to check your work?





Congrats! You have now mastered Class I and Class II cavities!

Contact Us!

If you have any further questions, stuck in any step or want to show us your progress, feel free to contact us completely free of charge!

Our talented instructor Dr. Salma is waiting for you!



Want to book a one on one session? Just give us a DM!

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Whatsapp: +201105723303



Thank you

Good Luck!

